Controllable Generative Design of Biological Sequences

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ML in PL
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HELMHOLTZ MUNICH

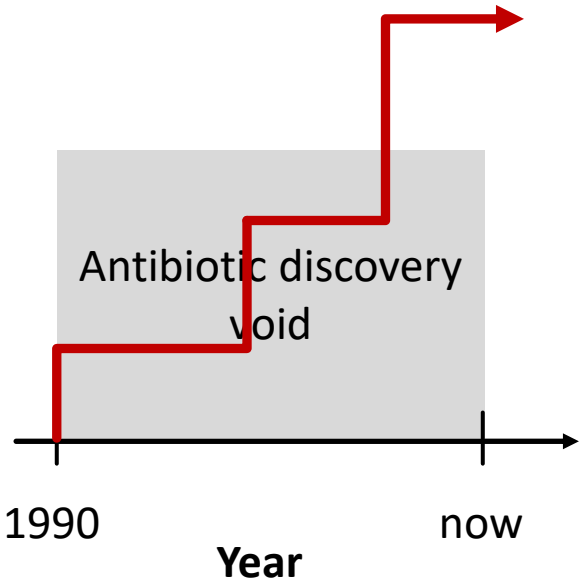
UNIVERSITY OF WARSAW

LMU LUDWIG-MAXIMILIANS-UNIVERSITÄT MÜNCHEN

Wait, why do we want to design biological sequences?

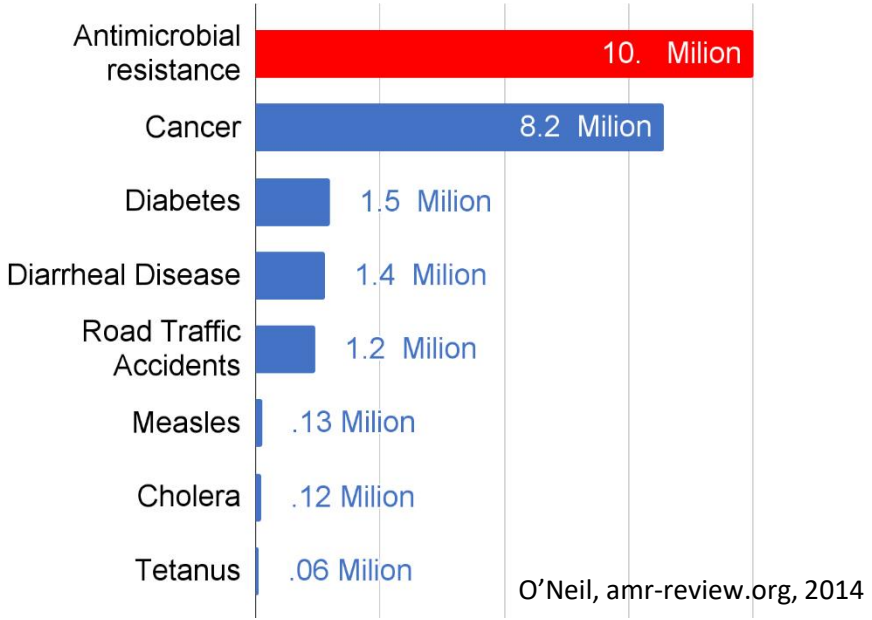
Example: to solve the problem of antimicrobial resistance

Resistance to existing antibiotics



Miethke et al, Nat Rev Chem, 2021

Projected no. of deaths per 2050 (global)



O'Neil, amr-review.org, 2014

Antimicrobial Resistance Collaborators, *The Lancet*, 2022

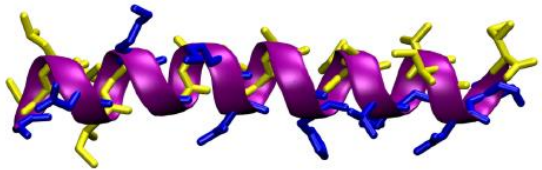
https://www.who.int/campaigns/world-antimicrobial-awareness-week/2022

Promising solution: antimicrobial peptides (AMPs)

Sequences like this: `GVLDILKGAAKDLAGHVATKVINKI`

Alphabet: 20 amino-acids

cationic hydrophilic

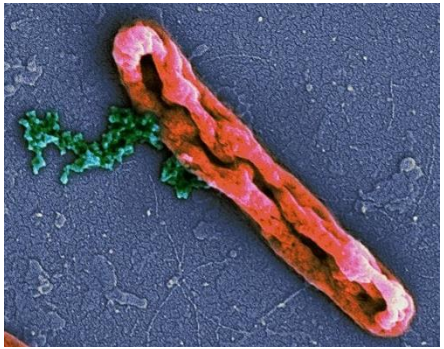


hydrophobic

Key physicochemical properties:

- High charge
- High hydrophobicity
- Adequate length (2-50aa)

Key mode of action:
membrane permeability



Key advantages over antibiotics:

- Specific for bacteria (non-toxic)
- Slower in inducing resistance!

The data about peptides: sequences plus (scarce) labels

Many peptide sequences (3.5 mln)

GIGKFLHSAKKFGKAFVGEIMNS
LLGDFFRKSKEKIGKEFKRIVQRIKDFLRNLVPRTES

GIGAILKVLATGLPTLISWIKNKRKQ
KWKLFKKIEKVGQNIRDGIIKAGPAVAVVGQATQIAK

ILPWKWPWWPWRR

ITSISLCTPGCKTGALMGCNMKTATCHCSIHVSK

ACYCRIPACIAGERRYGTCIYQGRLWAFCC
SSLLEKGLDGAKKAVGGLGALGKVGAEGLGVAEGIVDIVDSIKDVFG
DSSDPEV

FLPLIGRVLSGIL

GSKKPVPIIYCNRRTGKCQRM

TRSSRAGLQFPVGRVHRLLRK

GLFDIIKKIAESF

….

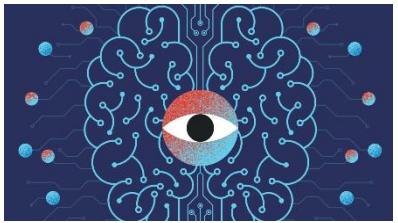
Few activity labels (MIC; ~5 K):

- Binary: AMP/non AMP
- Minimum inhibitory concentration against target bacteria, e.g.
  16µM against *E. coli*

Promising idea: AI-driven AMP design

Objective: generative AI models for AMP design!

Generative models

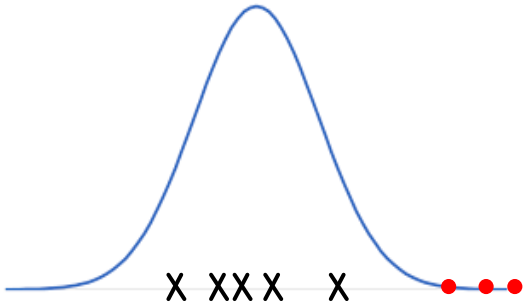


Design

GVLDILKGAAKDLAGHVATKVINKI

But: sing generative AI to design AMPs is difficult!

Data distribution learnt by generative model

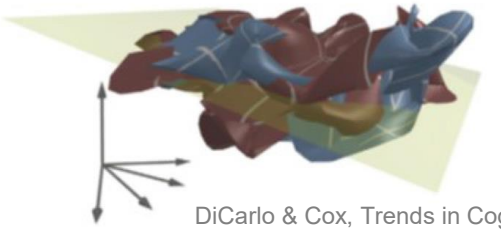


Likely samples (**realism**)

Samples with desired property/fitness (**idealism**)

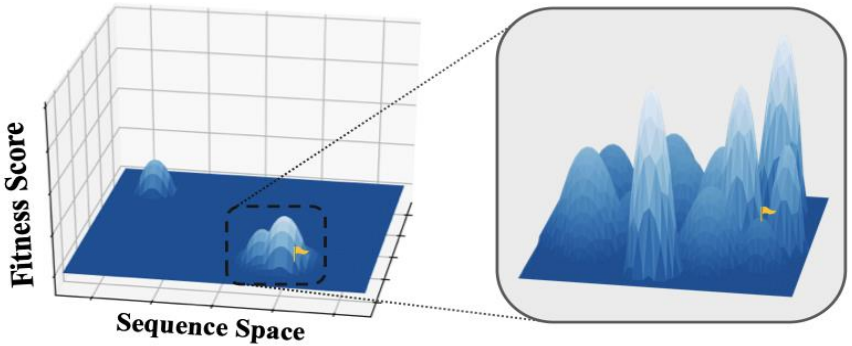
Idealism-realism tradeoff

- It is not about sampling **any** peptides
- But **navigable, controlled, optimized generation**
- Applies to any molecules



DiCarlo & Cox, Trends in Cognitive Sciences, 2007

Difficult to navigate data manifold

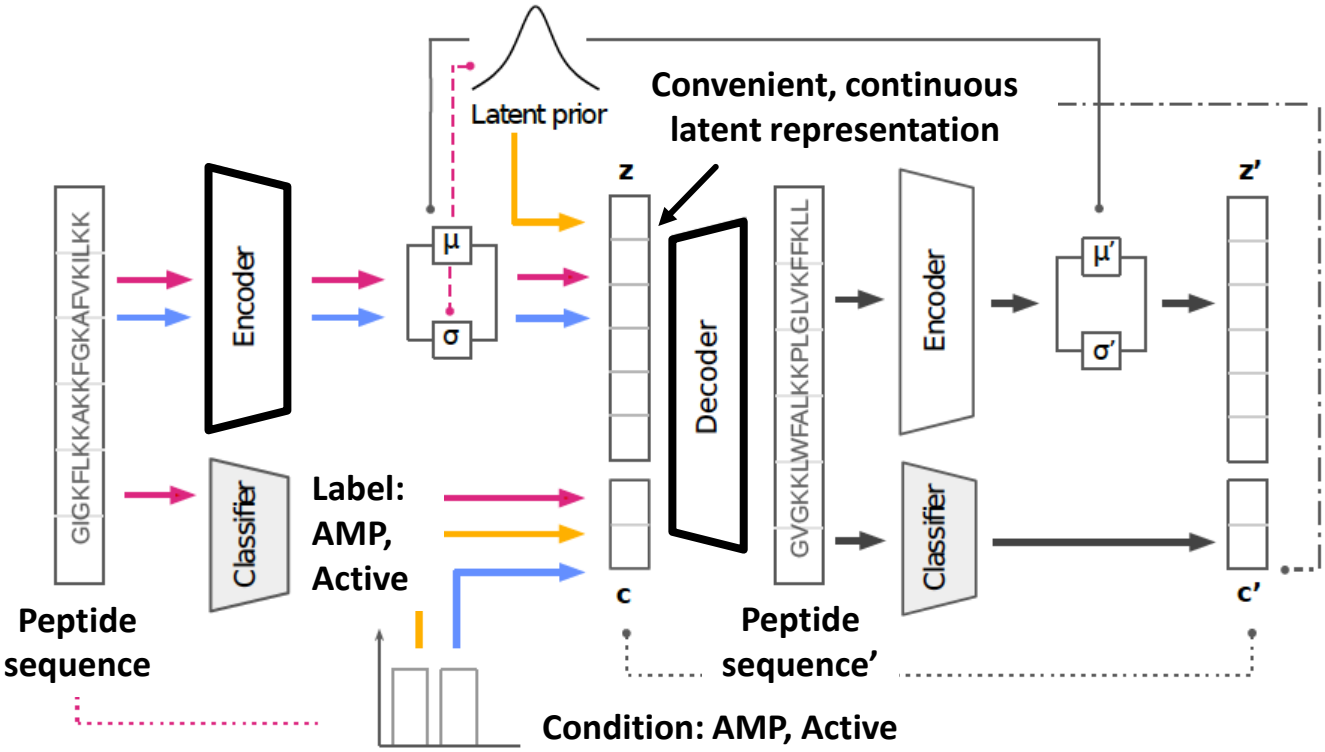


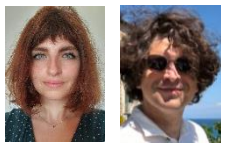
Ren & Li et al., ICML, 2022

Sparse and **OOD** ”fitness landscape”

Our first solution: generative model HydrAMP

Extended conditional variational autoencoder (VAE)



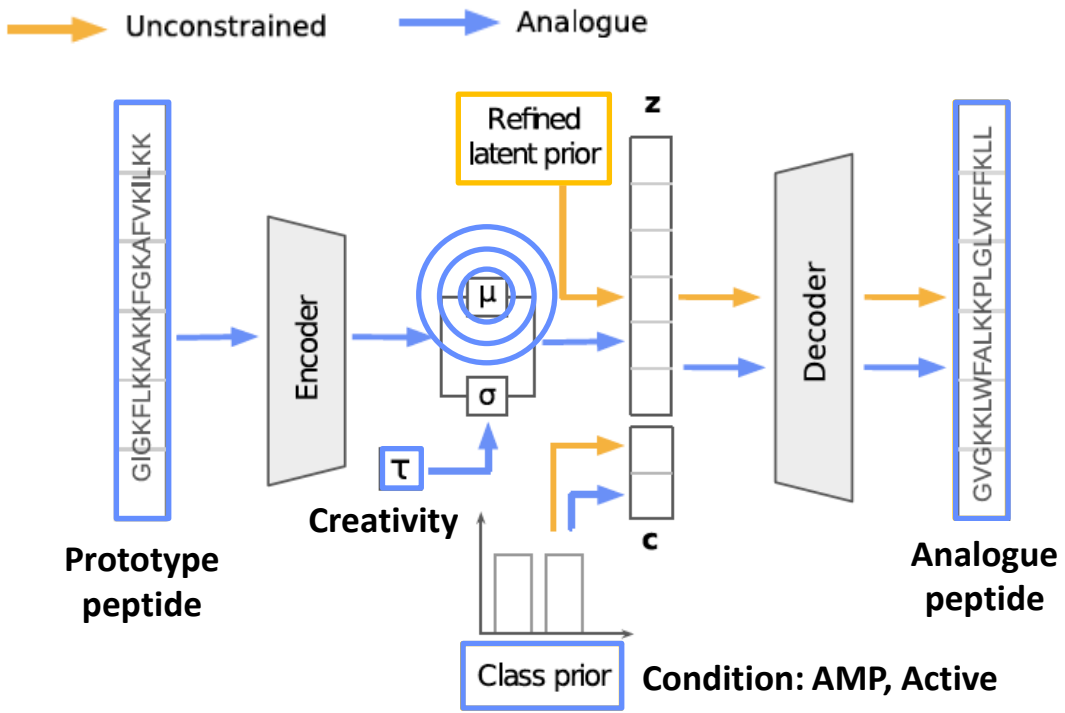


Szymczak, Możejko, *et al.*, *Nat Commun* (2023)

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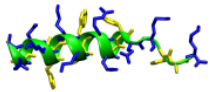
HydrAMP can generate in two ways

2 generation modes

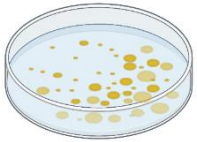


HydrAMP increased the success rate of experiments

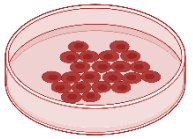




Peptide
synthesis



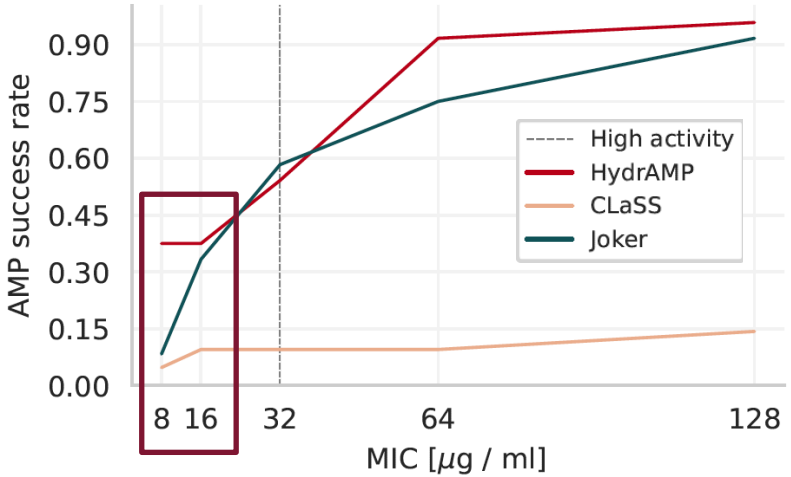
Microbiological assays
(**activity**; MIC; 5 strains)



Hemolytic activity assays
(**toxicity**; HC50)

- **Improved 4** active AMPs (Pexiganan, Syphaxin, OP-145, Omiganan) into **8 ones**
- **Turned 1 inactive** peptide (GQ20) into **7 active**
- Altogether **15 novel peptides:**
  - Active against *E. coli*, *S. aureus*, *A. baumannii* & *P. aeruginosa*
  - 12 Non-toxic

Increased success rate



Could name an AMP after my cat

Rudyxin

GRLLKFIKKAQRKFVEKFAK



Lets **improve navigability, functionality** and **controllability**!





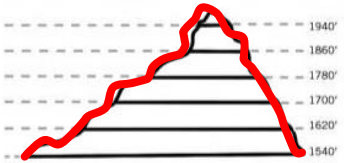


Lets look at the latent space as a map

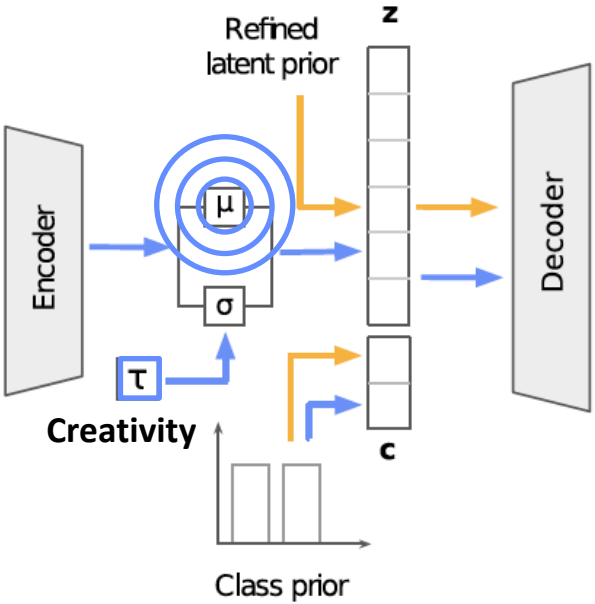
Latent space is a map



Straight lines are not the shortest

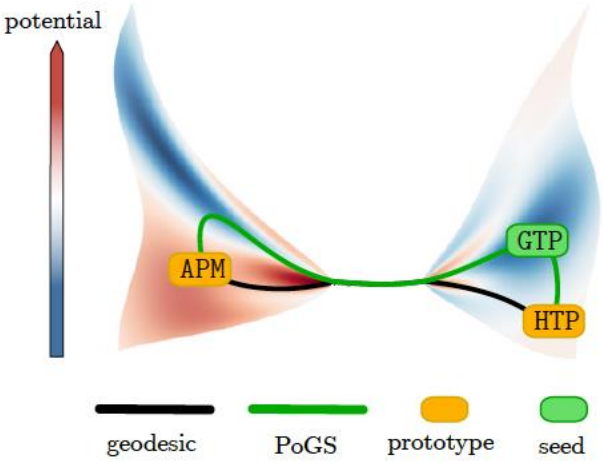


Inefficient “flat” search



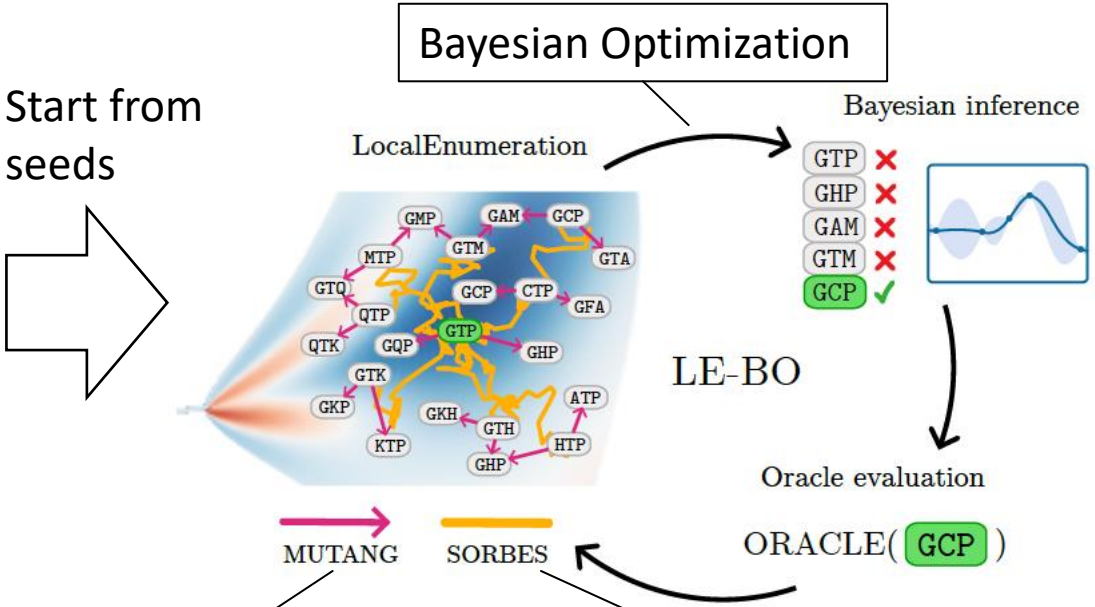
- Euclidian → Riemannian geometry
- Typical: Use a decoder-induced pullback metric
- **Does not work for peptides!** Varying intrinsic dimensionality

PepCompass: navigating the peptide space



Moving along geodesics maximizing predicted peptide activity

Start from seeds



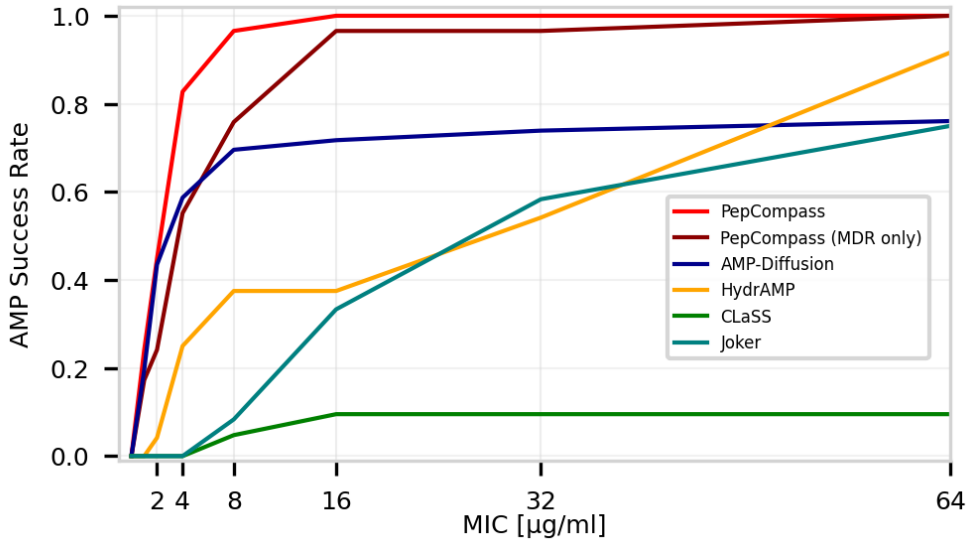
Mutation Enumeration in Tangent Space

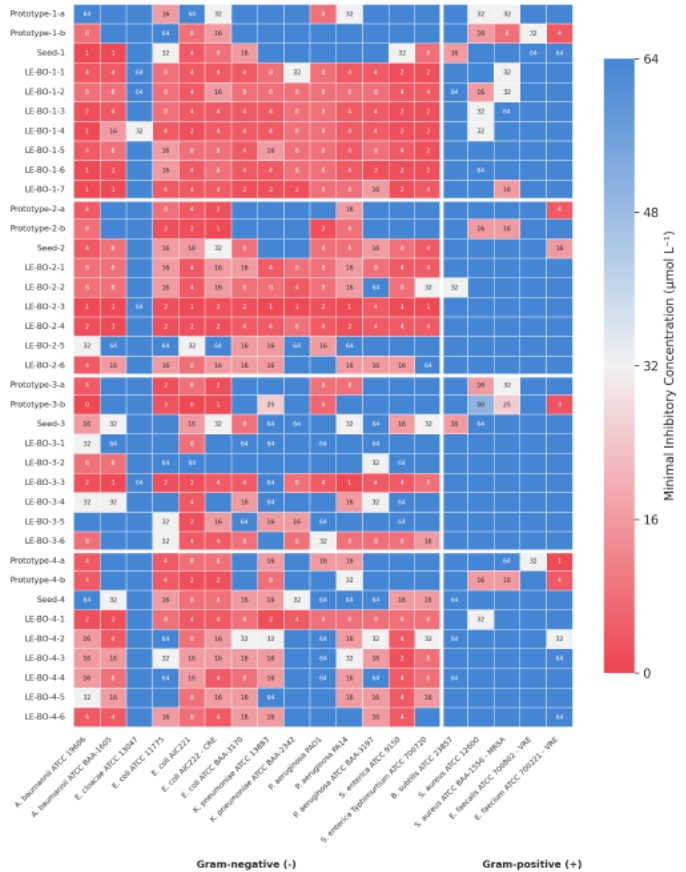
Second-Order Riemannian Brownian efficient Sampling



Możejko, Bielecki, *et al.*, *ICML* 2026

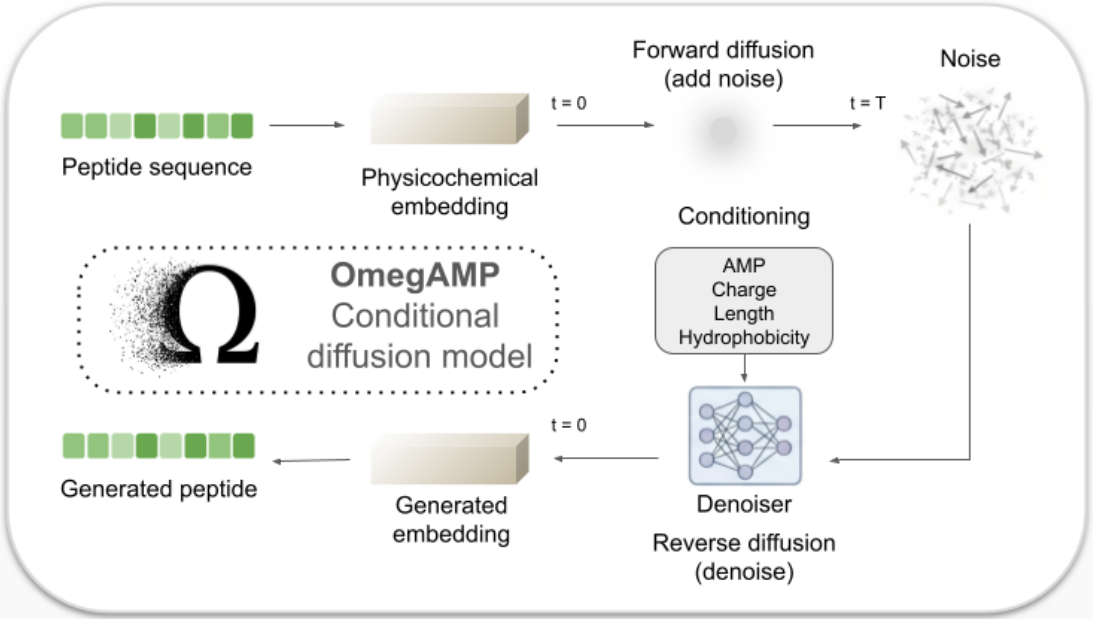
PepCompass designed AMPs with 100% success rate

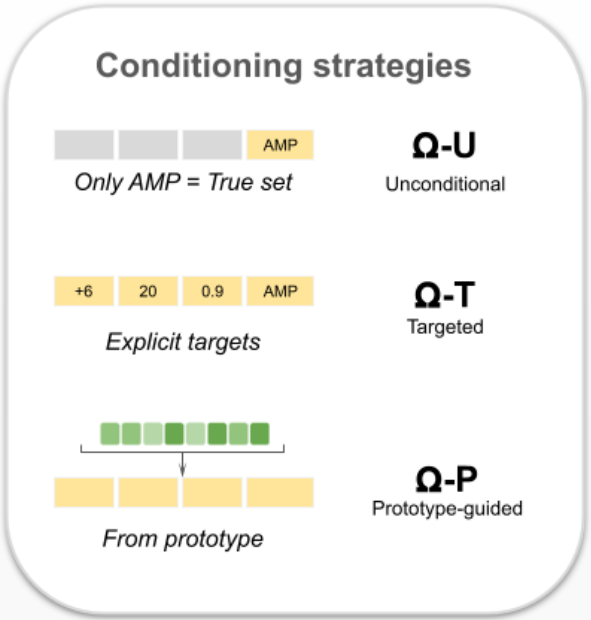


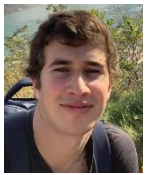


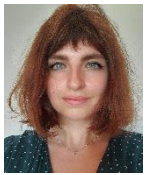
- Tested against 19 strains
- **Four active** AMP seeds
- **25/25 total active AMPs**

OmegAMP: a conditional and guided AMP diffusion model



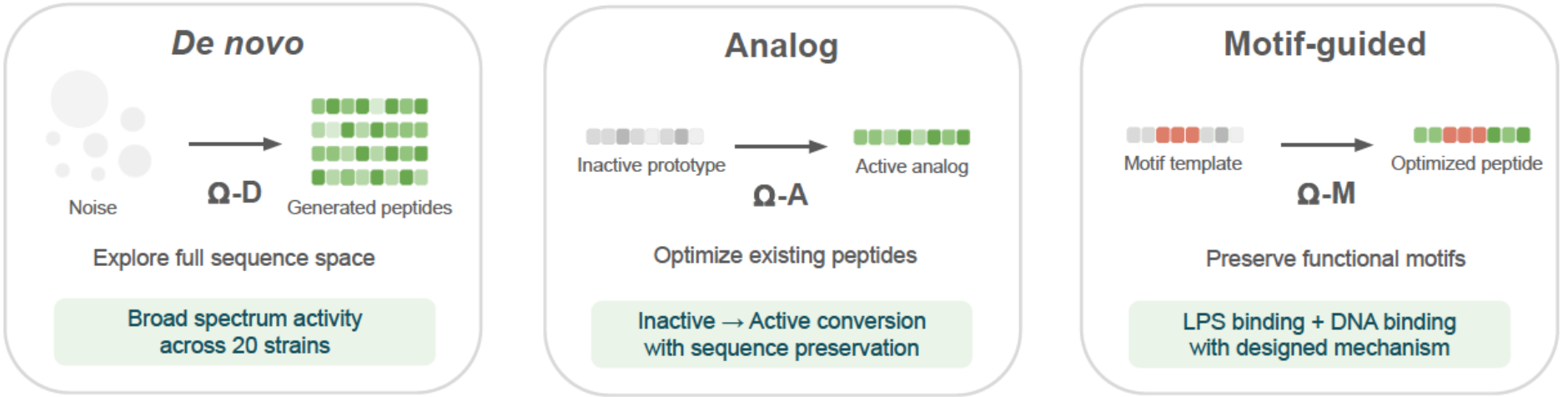




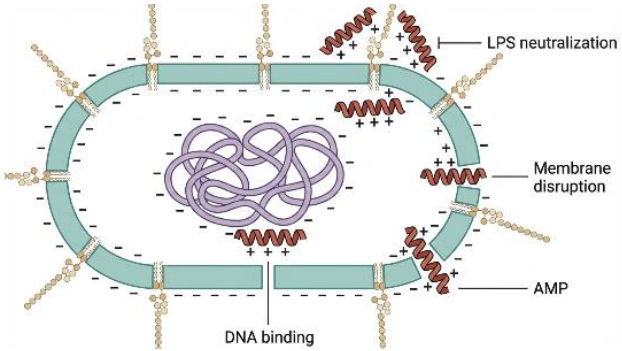


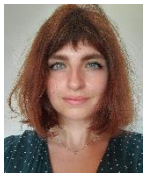
Soares, Hetzel, Szymczak, et al *TMLR,* 2026

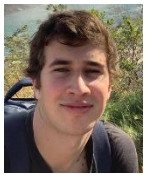
OmegAMP: generation modes



Hypothesis:
Analogues + motifs enable design of AMPs with controllable mechanism of action, safety and novel functions



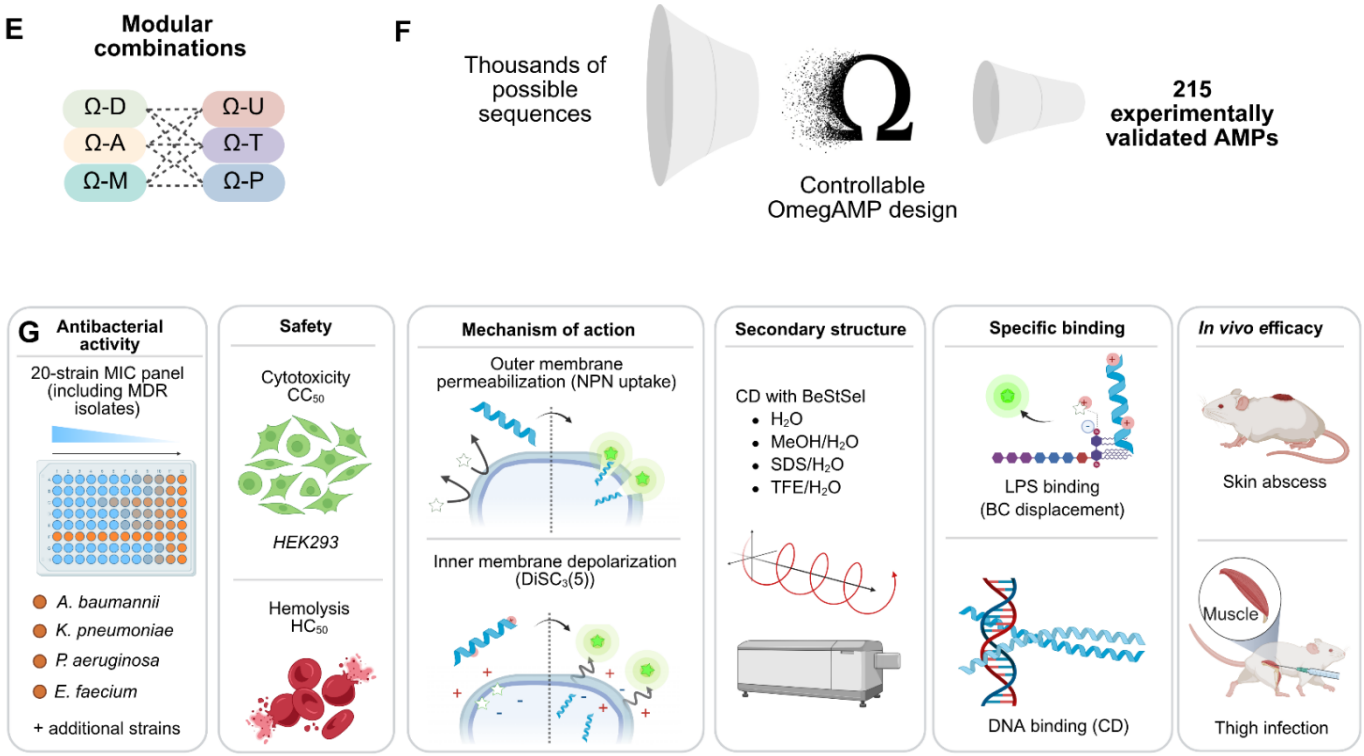




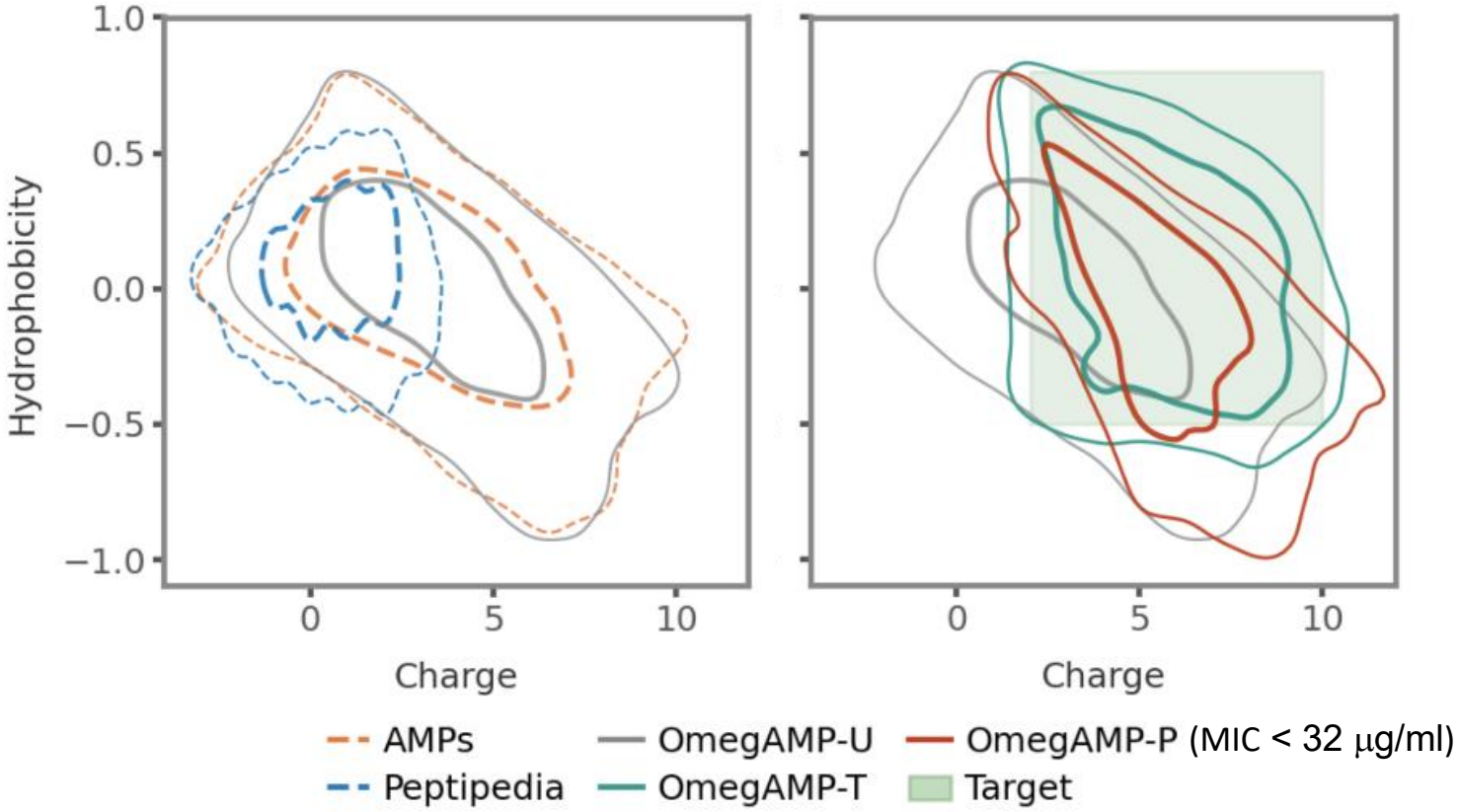

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Szymczak, D.T. Torres, Soares, Hetzel, et al *Biorxiv,* 2026

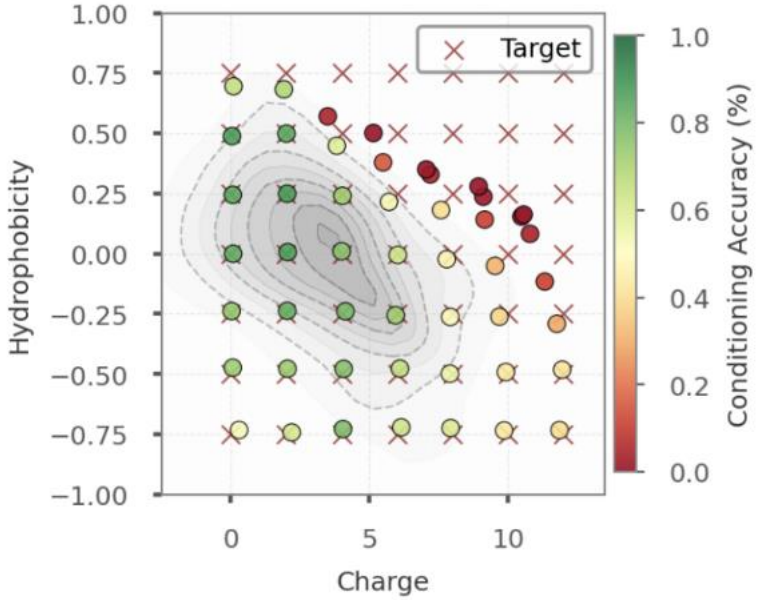
Extensive experimental validation

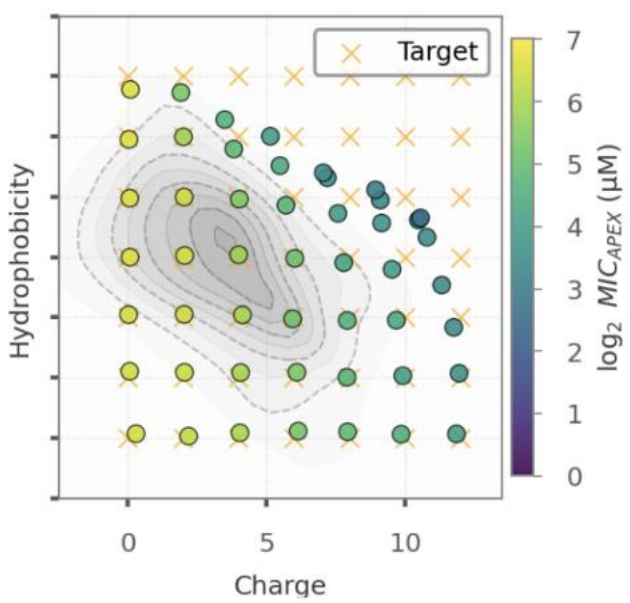


Conditioning causes expected data distribution shifts

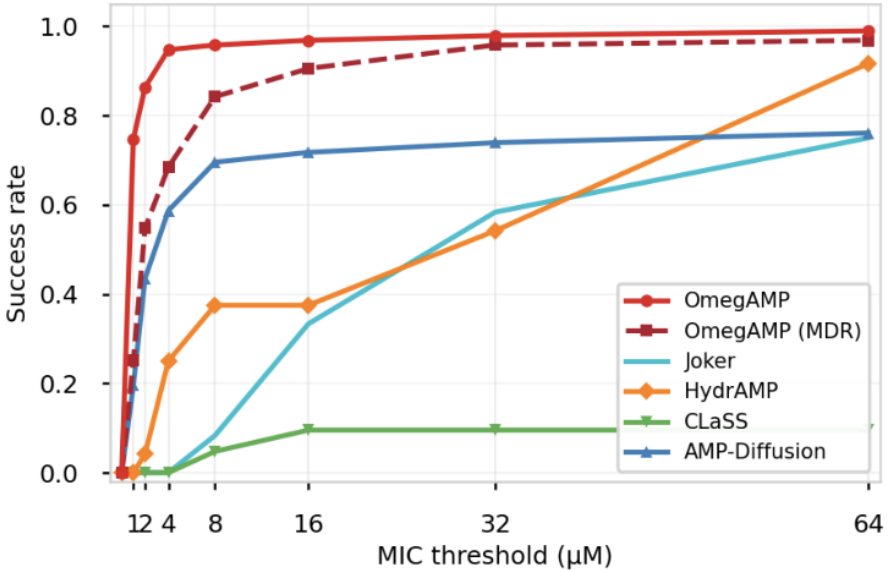


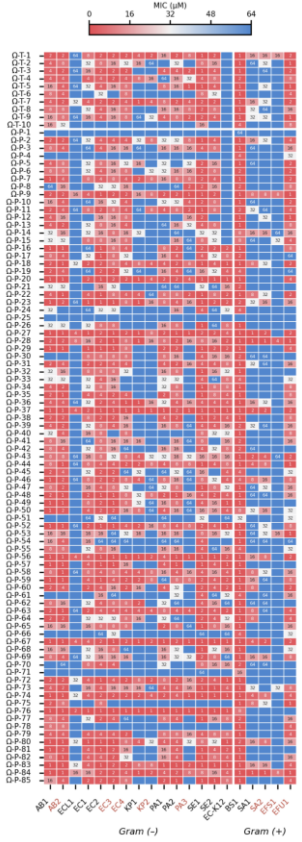
Targeted conditioning reveals learned physicochemical constraints



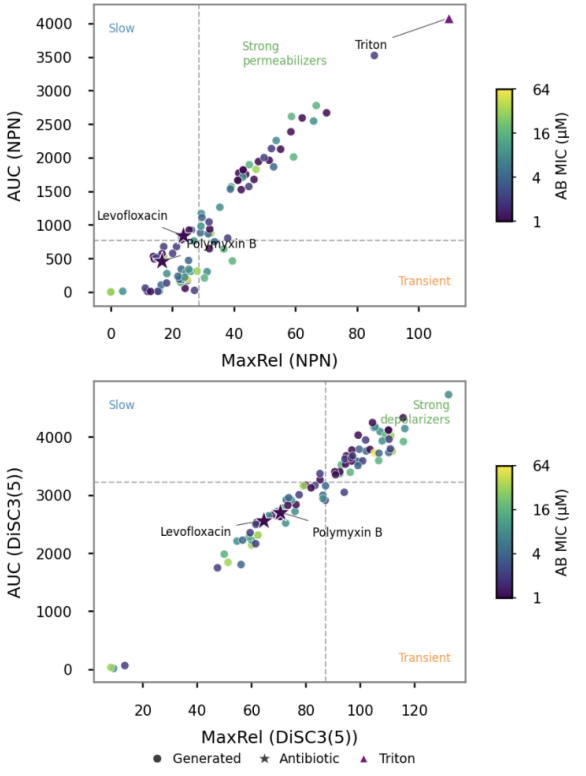


OmegAMP-D also rocks in terms of experimental validation!



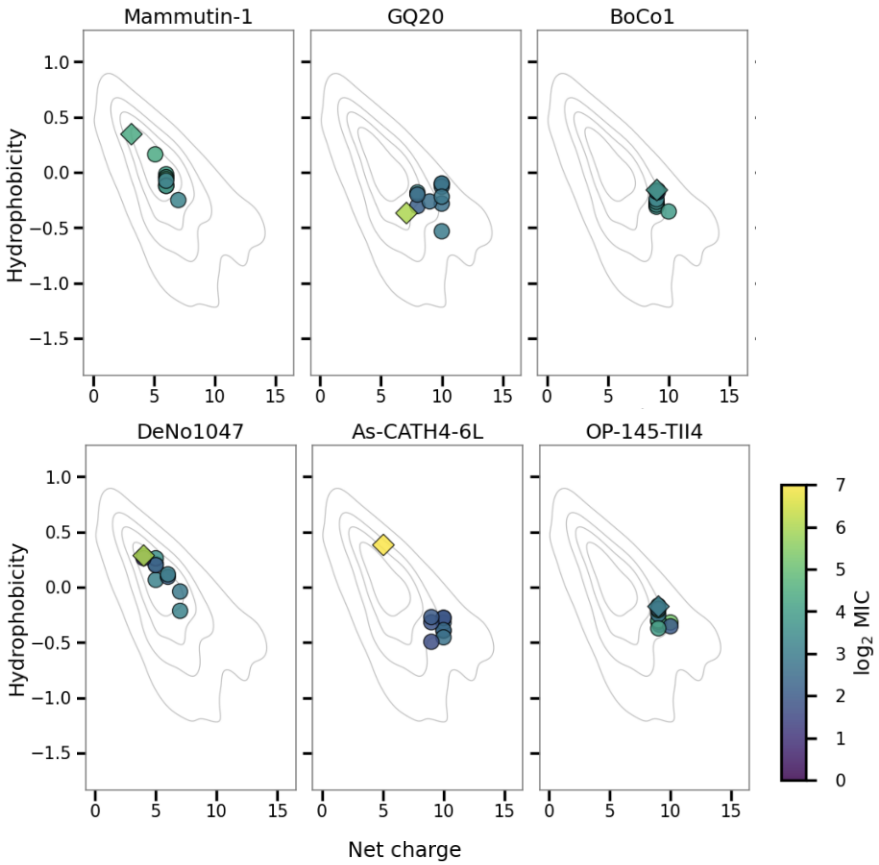


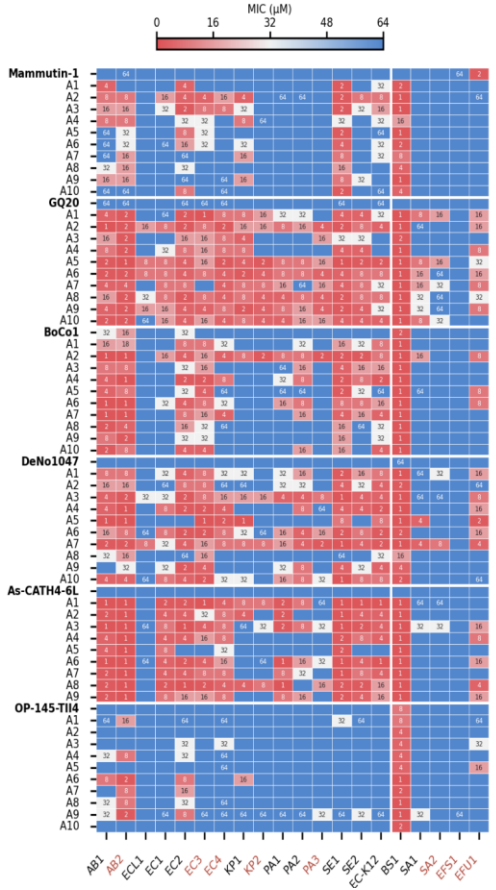
~50% membrane-active



- 10 peptides by OmegAMP-T
- 85 peptides by OmegAMP-P
- 93 (98%) exhibited MIC ≤ 32 μM against at least one strain

Analogues designed by OmegAMP-A depend on the prototypes

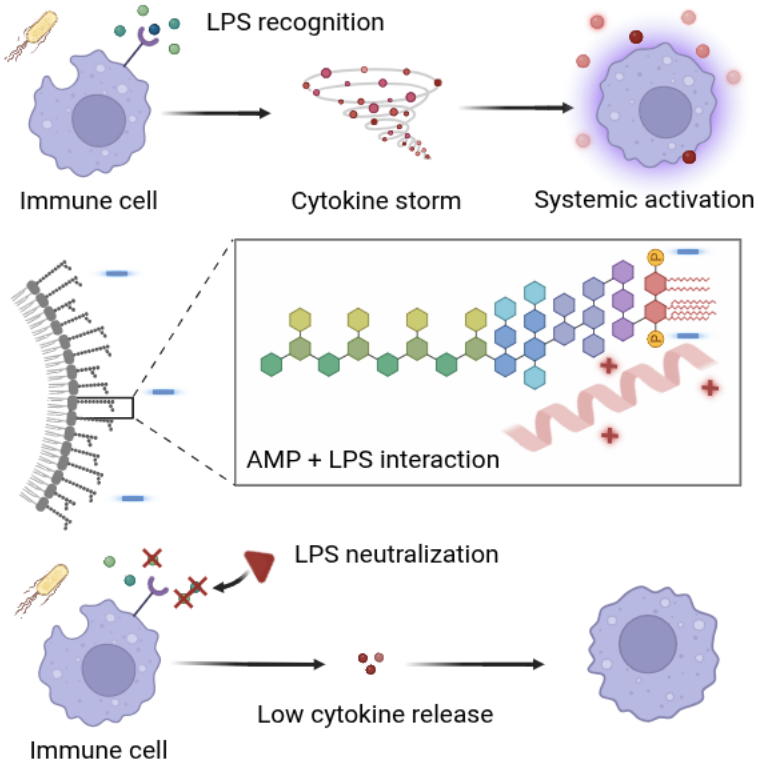


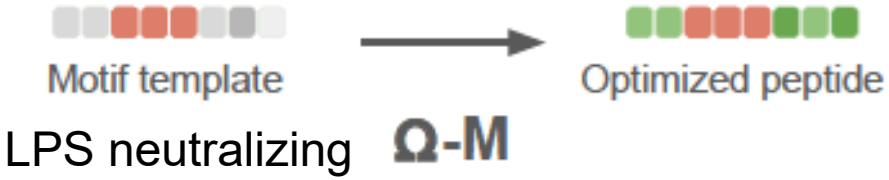


Per prototype success rate related to sequence

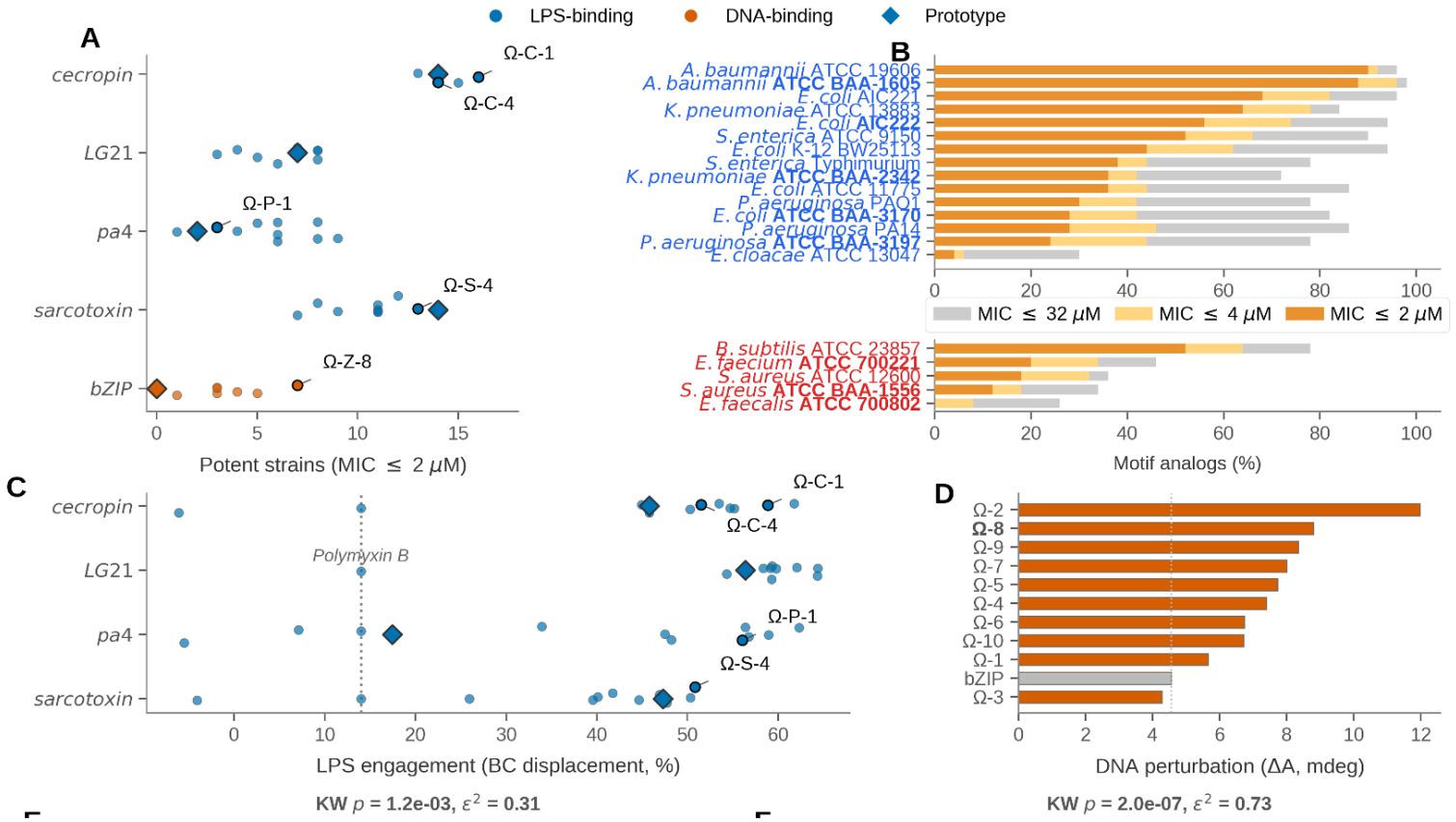
Motif+analog OmegAMP generation designs LPS-binding AMPs

Lipopolysaccharide neutralization

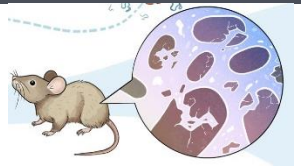


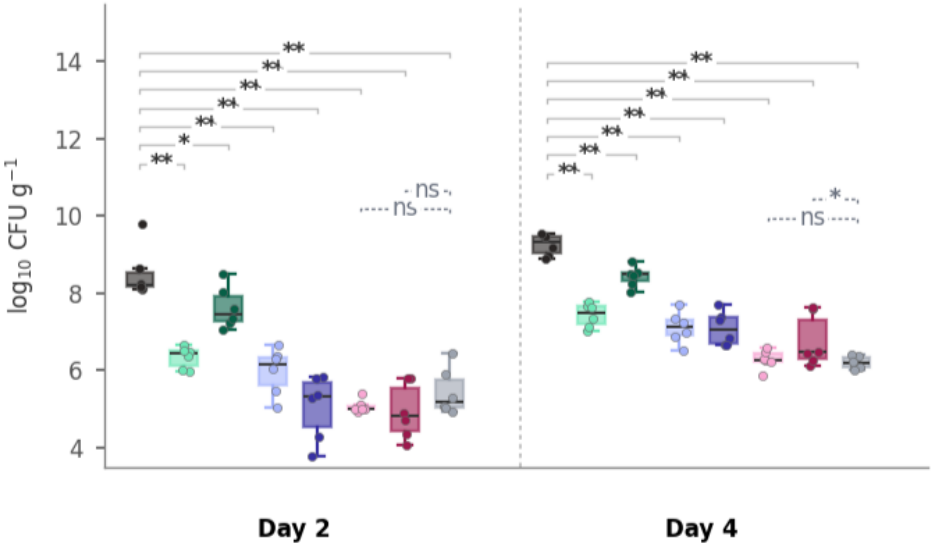


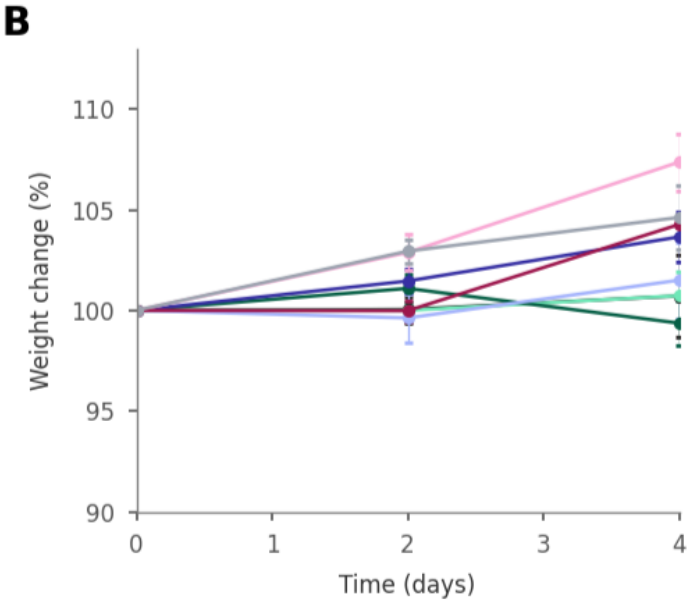
AMPs that succeed in LPS and DNA binding



Experiments on mice confirm activity in vivo

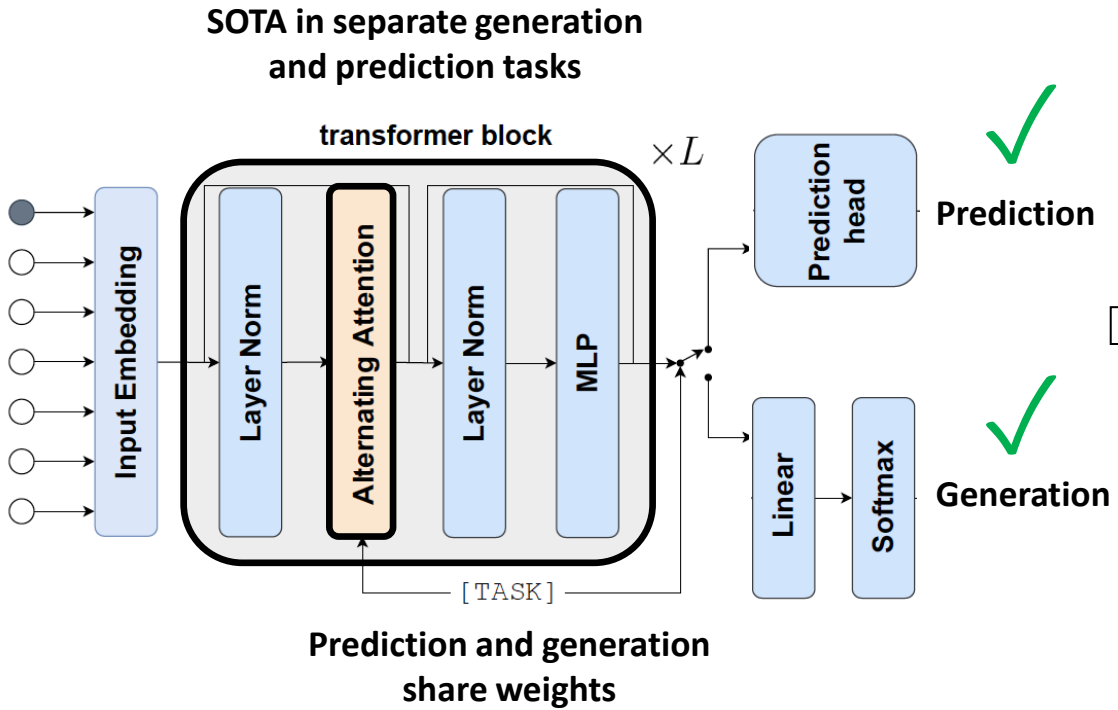






Control
Ω-AT-BoCo1-5
Ω-AT-BoCo1-9
Ω-DP-52
Ω-DP-19
Ω-AMT-cecropin-1
Ω-AMT-cecropin-4
Polymyxin B

HyFormer: generation and prediction using a transformer



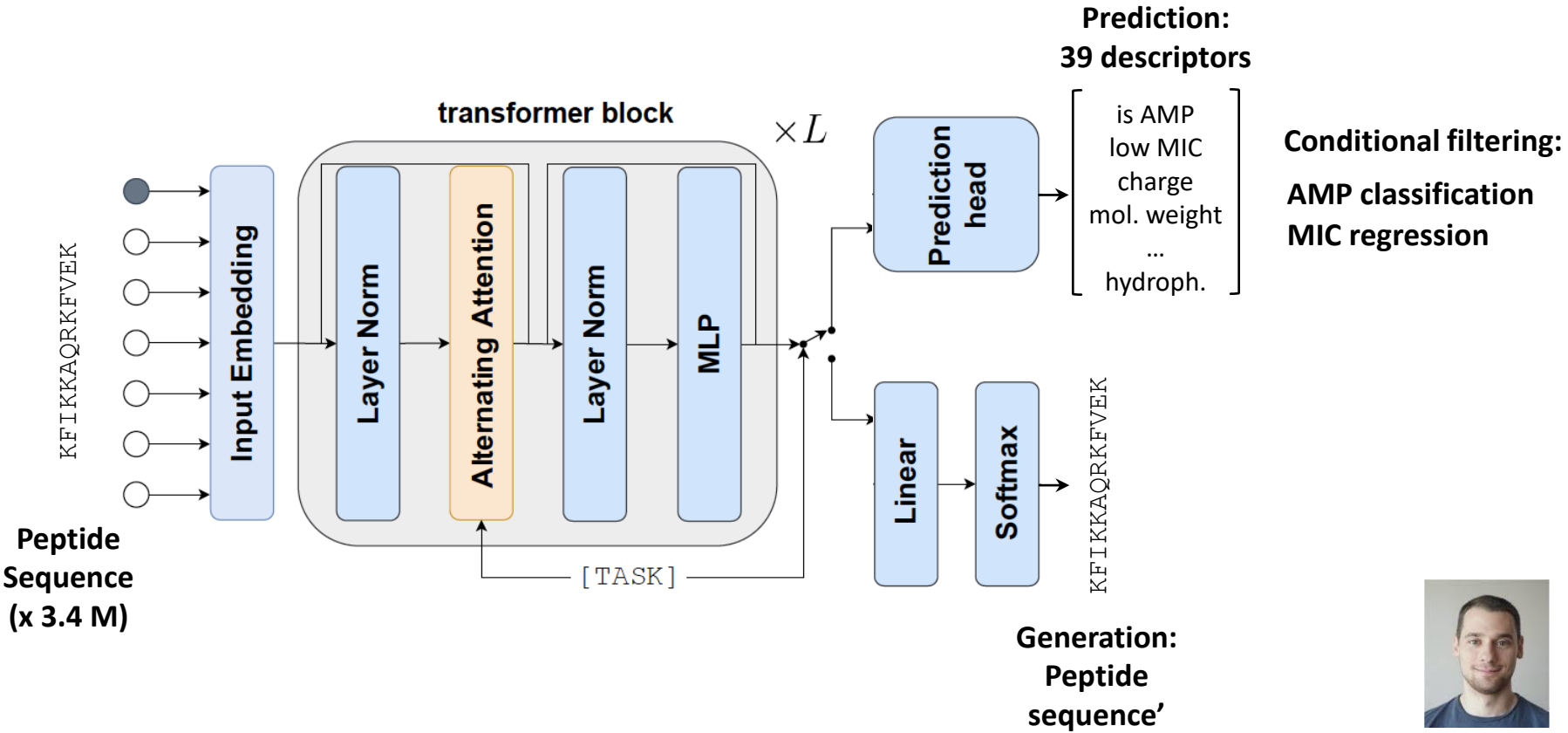
Synergistic benefits

- Representation learning
- Generalization to OOD settings
- Conditional sampling



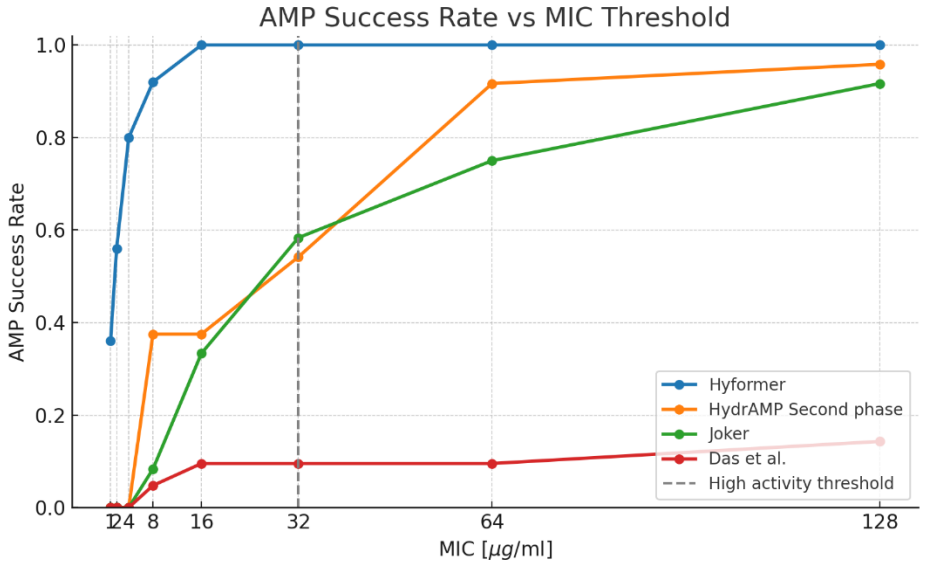
Izdebski, *et al.*, *TMLR*, 2026
EDS spotlight, J2C certification, presented at ICLR 2026

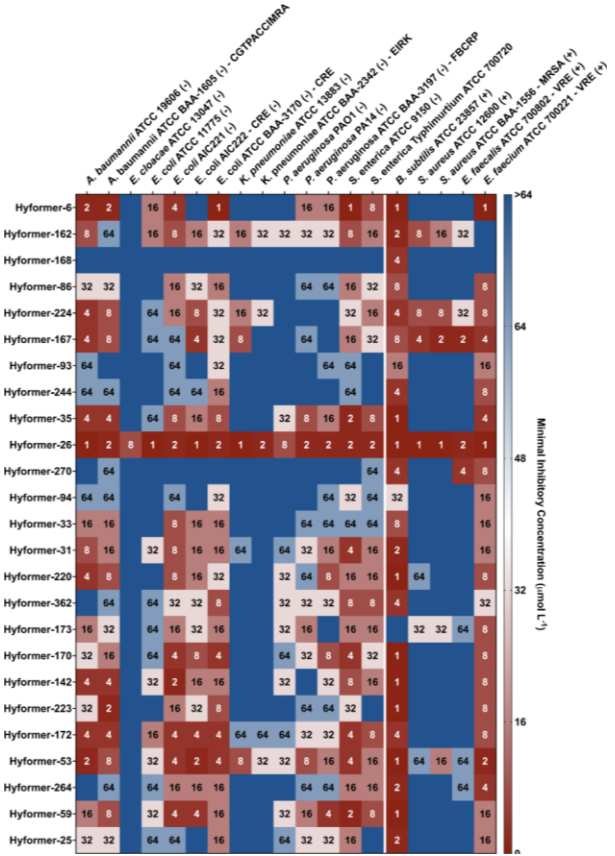
HyFormer as conditional generator for peptides



Izdebski, *et al.*, *TMLR*, 2026
EDS spotlight, J2C certification, presented at ICLR 2026

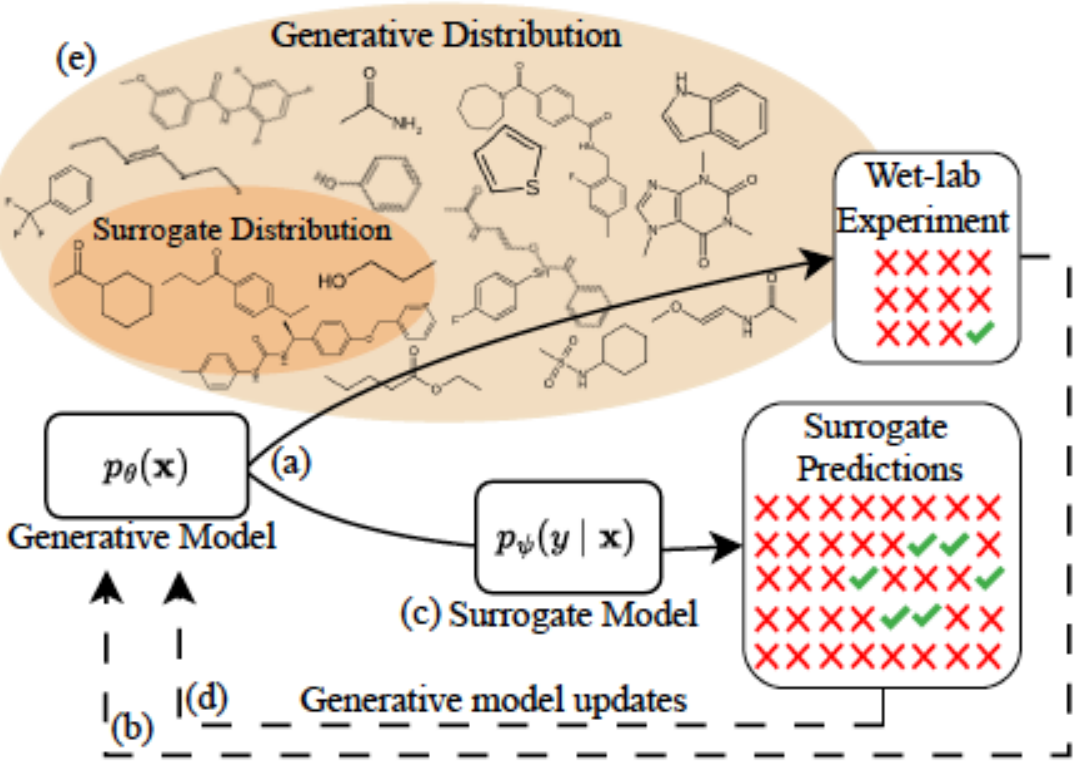
HYFORMER also achieved ~100% experimental success rate

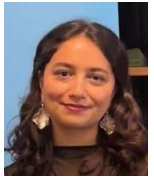




- Tested against 19 strains
- 100% success rate with MIC = 16
- 1 peptide active against all tested bacterial strains

Iterative optimization: issues with out of distribution and variance

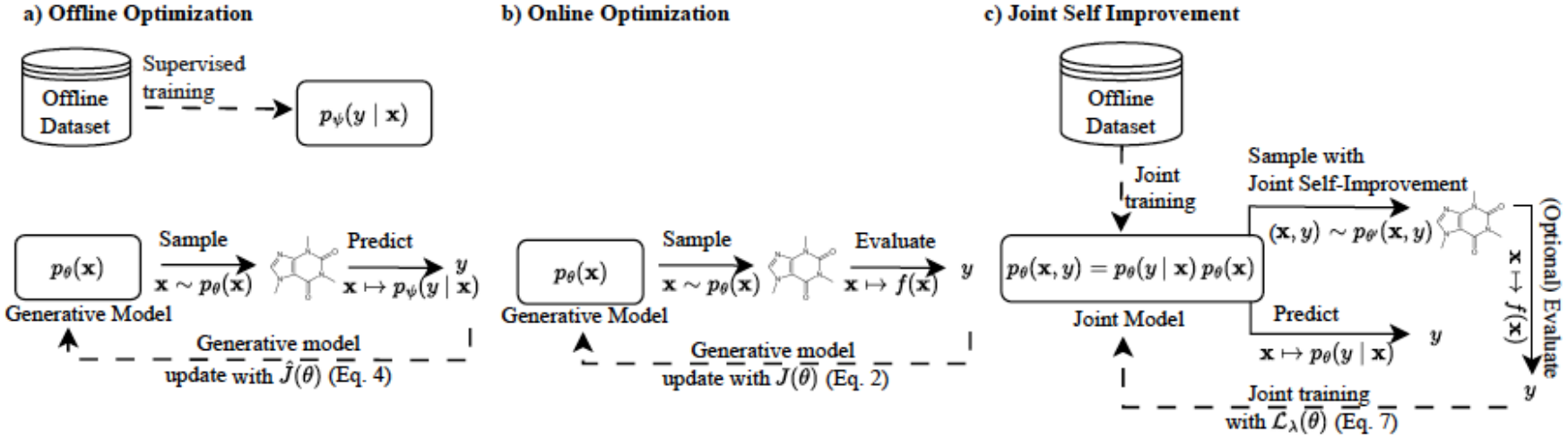




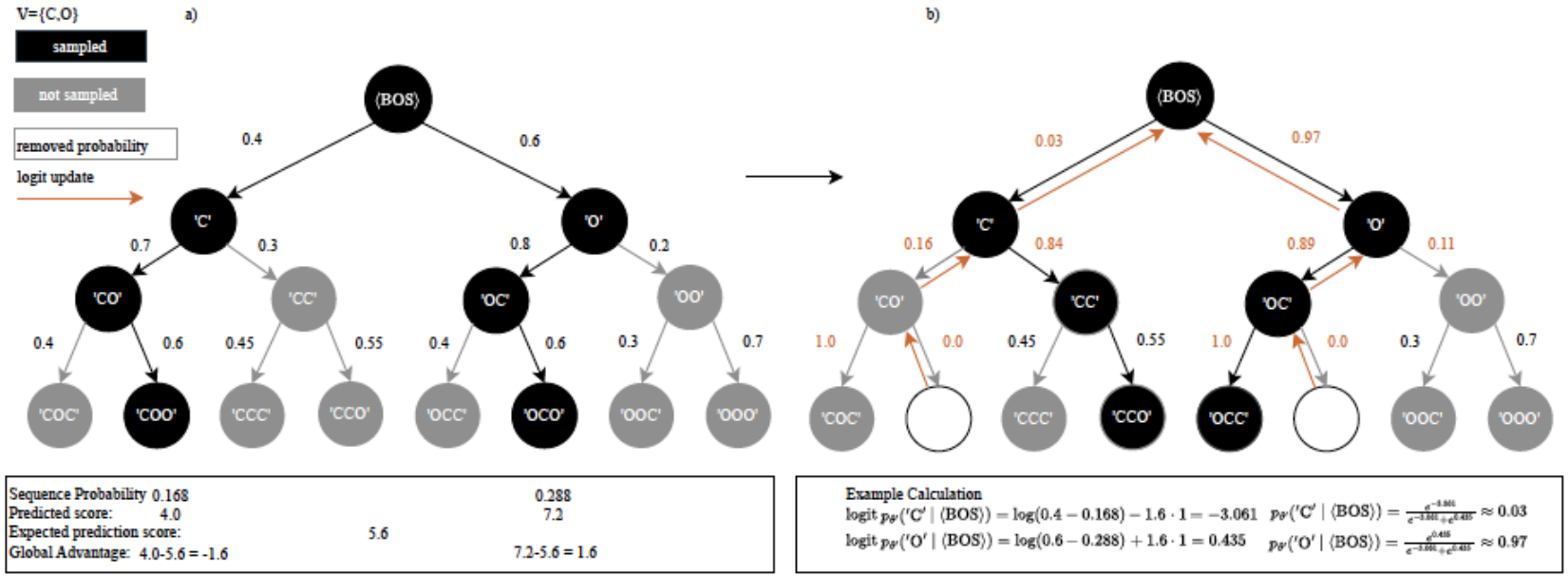


Korkmaz, Izdebski, *et al.*, *NeurIPS,* 2026

Joint self improvement works both in online and offline setting



Biasing inference time distribution toward high award samples



Sample efficiency is what matters when the data is scarce

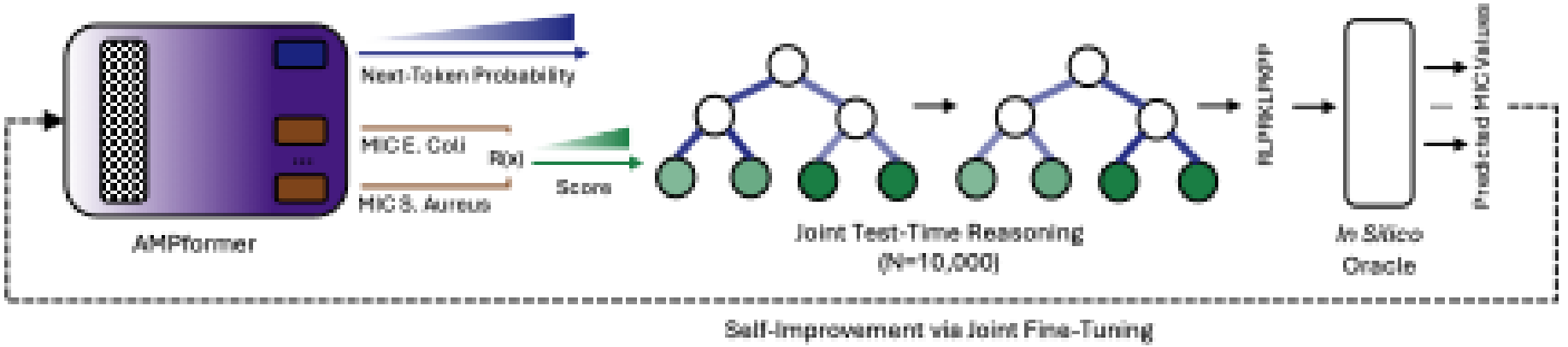
Table 4: Molecular optimization performance on offline AMP optimization. Best method is **bold**.

MODEL	PERPLEXITY	DIVERSITY (↑)	FITNESS (↑)	HYDRAMP$_{MIC}$ (↑)	AMPLIFY (↑)	AMPEPPY (↑)
PEPCVAE	20.39 (20.17, 20.64)	**0.87 (0.87, 0.87)**	0.07 (0.07, 0.07)	0.12 (0.10, 0.14)	0.48 (0.46, 0.50)	0.52 (0.51, 0.53)
AMPGAN	18.97 (18.76, 19.19)	0.80 (0.80, 0.81)	0.12 (0.12, 0.13)	0.22 (0.20, 0.25)	0.66 (0.63, 0.68)	0.55 (0.54, 0.57)
HYDRAMP	19.98 (19.71, 20.27)	0.86 (0.86, 0.86)	0.09 (0.08, 0.09)	0.44 (0.41, 0.47)	0.60 (0.57, 0.62)	0.52 (0.52, 0.53)
AMP-DIFFUSION	18.33 (18.03, 18.64)	0.82 (0.82, 0.82)	0.11 (0.10, 0.11)	0.18 (0.16, 0.21)	0.14 (0.12, 0.15)	0.33 (0.31, 0.34)
HYFORMER	17.72 (17.47, 17.97)	0.74 (0.73, 0.74)	**0.21 (0.20, 0.22)**	0.78 (0.76, 0.81)	0.96 (0.95, 0.97)	0.71 (0.70, 0.72)
JSI (OURS)	15.93 (15.68, 16.17)	0.78 (0.78, 0.79)	0.12 (0.12, 0.12)	**0.84 (0.82, 0.87)**	**0.98 (0.97, 0.98)**	**0.78 (0.77, 0.78)**

Table 5: Molecular optimization performance on offline AMP optimization with restricted proportion of hits. Best method is **bold**.

MODEL	PERPLEXITY	DIVERSITY (↑)	FITNESS (↑)	HYDRAMP$_{MIC}$ (↑)	AMPLIFY (↑)	AMPEPPY (↑)
1% HITS						
HYFORMER	16.84 (16.59, 17.09)	**0.83 (0.82, 0.83)**	0.11 (0.11, 0.12)	0.20 (0.18, 0.23)	0.76 (0.74, 0.78)	0.65 (0.64, 0.66)
JSI (OURS)	18.36 (17.94, 18.77)	0.69 (0.68, 0.70)	**0.16 (0.16, 0.17)**	**0.68 (0.65, 0.71)**	**0.98 (0.97, 0.98)**	**0.72 (0.72, 0.73)**
0% HITS						
HYFORMER	16.77 (16.51, 17.01)	**0.83 (0.83, 0.83)**	0.11 (0.10, 0.11)	0.17 (0.15, 0.19)	0.71 (0.69, 0.73)	0.62 (0.61, 0.63)
JSI (OURS)	18.34 (18.05, 18.66)	0.72 (0.72, 0.73)	**0.19 (0.18, 0.20)**	**0.66 (0.63, 0.69)**	**0.98 (0.97, 0.99)**	**0.71 (0.70, 0.72)**

AMPFormer works iteratively for peptide optimization

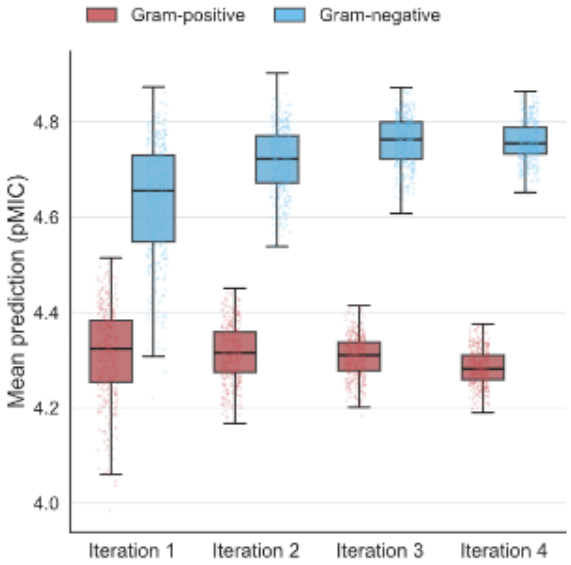


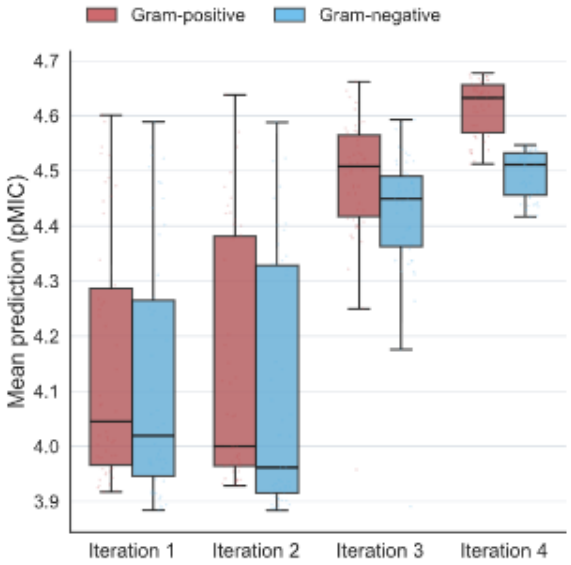


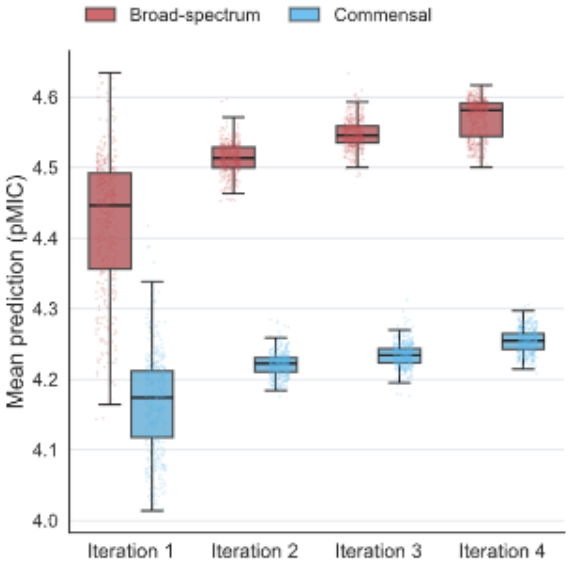


P. Gawade, A. Izdebski, Unpublished

Towards bacterial specificity







Achievements and challenges

Achievements

- Machine understanding of the ‘language of molecules’
- Generation of experimentally validated AMPs
- Multi-functionality: generation and prediction
- Multi-controllability: property, target, analogue and template
- Optimization via latent BO / active learning and guided SMC





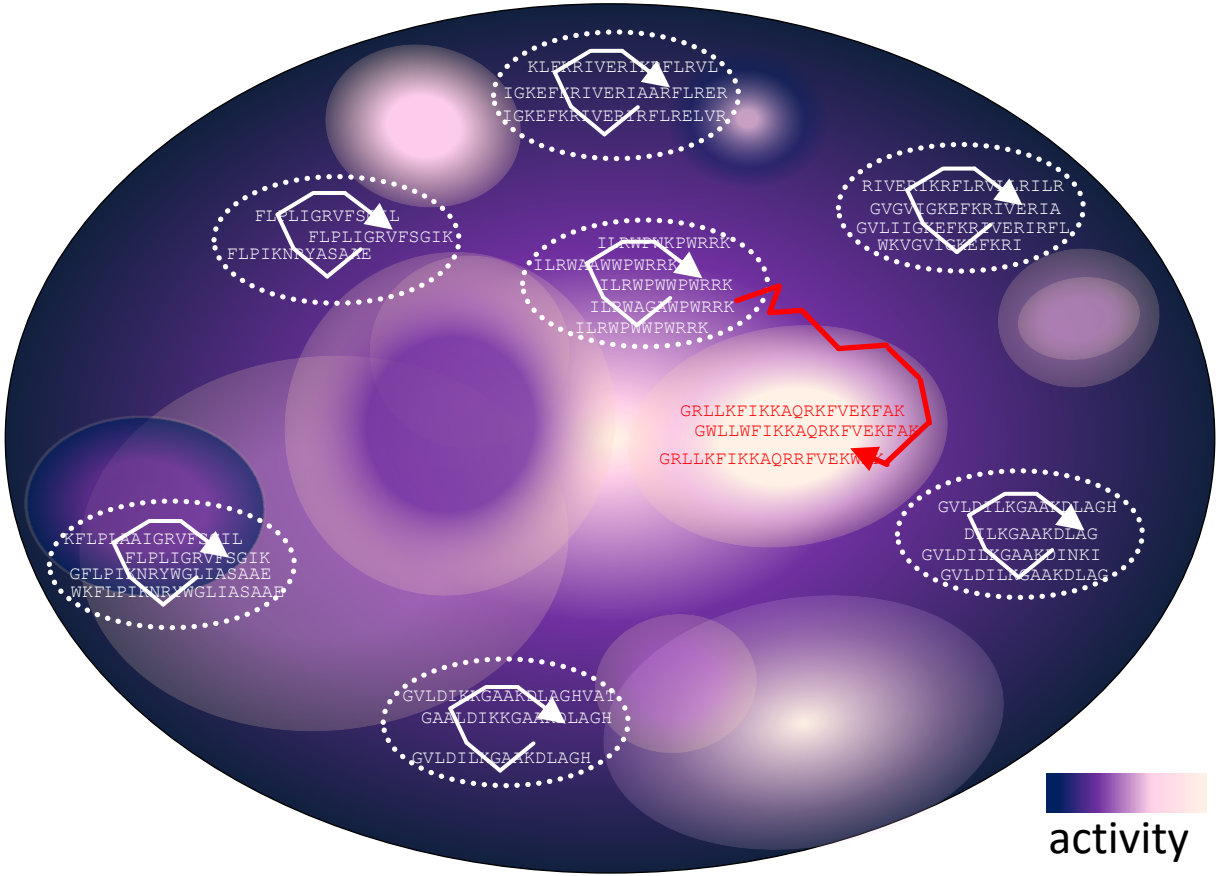
Challenges

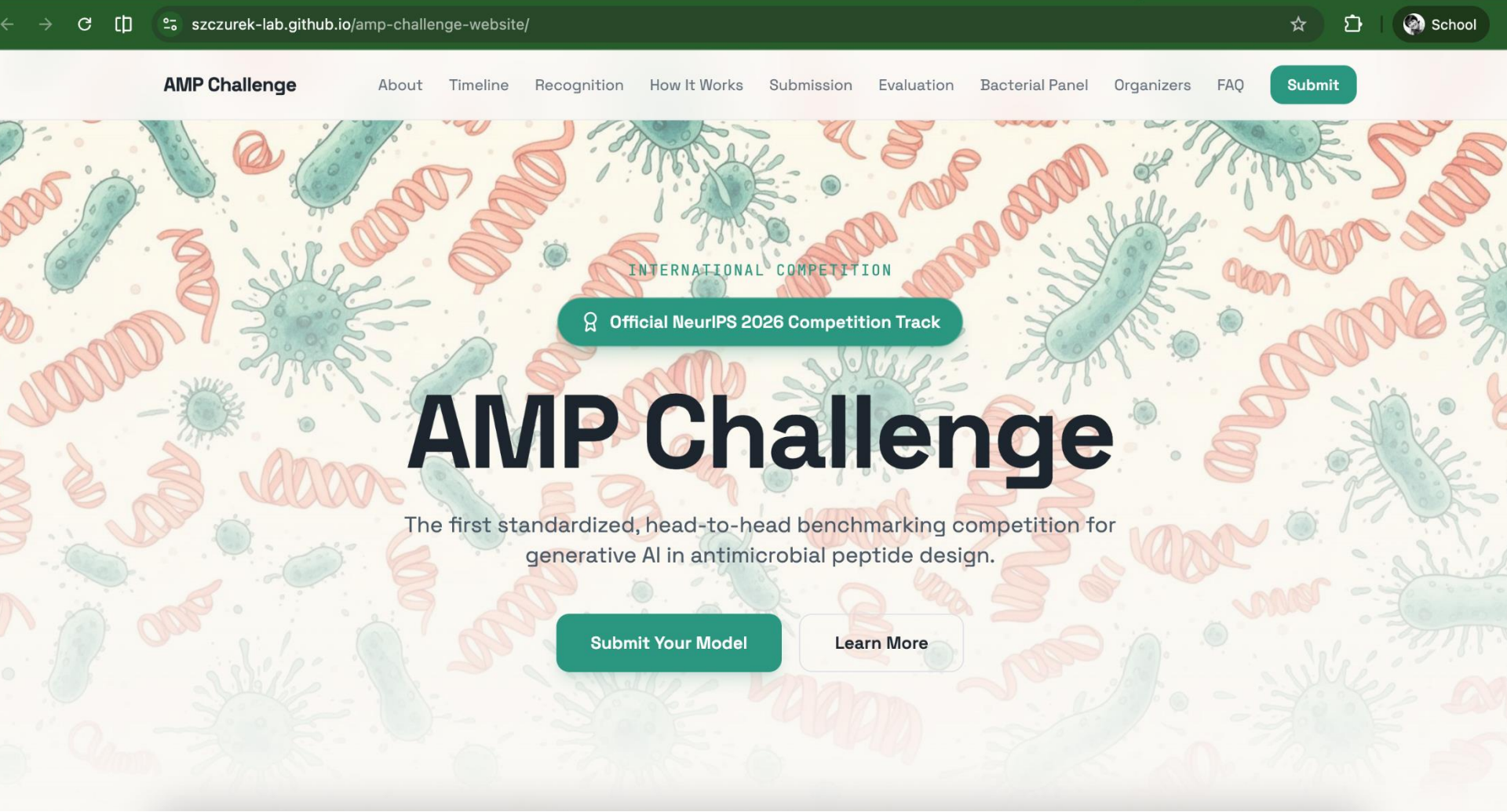
- Machine learning: lack of standard data and benchmarks
- Solved: active peptides; Open: mode of action & more
- Only linear peptides or lead molecules modeled
- For clinical applications need to be modified chemically

Szymczak *et al.*, *Acc. Chem. Res.*, *2025*
Szymczak & Szczurek, *Curr. Opin. Struct. Biol.*, *2024*

Why generative AI is such a paradigm shift

- Whole AMP space ~10^{32}
- Areas of different activity
- AMP data scarce
  250 K peptides, 20 K AMPs
- Groups of similar peptides
- Search around old AMPs
- Move to a better area
  of the AMP space




szczurek-lab.github.io/amp-challenge-website/
School
AMP Challenge
About
Timeline
Recognition
How It Works
Submission
Evaluation
Bacterial Panel
Organizers
FAQ
Submit
INTERNATIONAL COMPETITION
Official NeurIPS 2026 Competition Track
AMP Challenge
The first standardized, head-to-head benchmarking competition for generative AI in antimicrobial peptide design.
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Acknowledgement

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